

Candidates for a first-principles derivation of (r_e/r_0)

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Candidates for a first-principles derivation of r_e/r_0 — for Tepper Gill

Date: 2026-05-26 (revised 2026-05-26 with two waves of overnight progress on all three candidates). **From:** Trey Morris (with Claude Opus 4.7). **Re:** Follow-up to the 2026-05-26 triangulation note ([Roadmapping/Author_Reports/2026-05_re_triangularization_followup_for_gill.md](#)); Scope 1 of [issue #54](#). The *Progress update* section reports findings from three parallel research branches (issues [#64](#), [#65](#), [#66](#) under master [#67](#)). Two of the three branches have now reached definitive verdicts: **Candidate 2 halted at Outcome D** (the published §III.D NR-expansion is structurally inadequate to pin r_e at any scale where the expansion is valid), and **Candidate 3 halted at Outcome C-as-published** (the closed-form Schwinger match is *algebraically forced* by the back-fit definition of the triangulated cutoff, not evidence of intentional encoding). Candidate 1 is live, scaffolded, and on a clear next-step plan.

Context

The DRQM I §III.D r_e value was originally a *cutoff parameter*, fixed by a numerical search against the measured electron g_s . The empirical refinement in [PR #62](#) widened that search to a joint fit across six observables (electron g_s , H $2P_{3/2} - 2P_{1/2}$, H 1S hyperfine, He $^3P_0 - ^3P_1$, positronium ortho-para, muonium hyperfine), and returned $r_e/r_0 = 0.499\,420\,509\,912\,831\,7$. The honest reading of this result, recorded in detail in [Bethe_Salpeter/10_CrossComparison.md §2](#): the six observables are not six independent constraints on the cutoff, because each is computed as $(g_s/ - 2)^n \times \text{textbook}$ with $n \in \{1, 2\}$, so all six share a single dependence on g_s and therefore on r_e . The joint fit is *one structural fact* (the single-parameter cutoff prescription) applied to six manifestations, not six independent tests. What the multi-observable fit *does* confirm is that the six manifestations are mutually consistent at the framework's precision floor — i.e., the $(g_s/ - 2)^n \times \text{textbook}$ scaling holds across all six — which is a self-consistency check on the framework's structural prescription, not an independent corroboration of the cutoff's value.

There is, at present, **no first-principles derivation of r_e/r_0** anywhere in the campaign or in the published DRQM I material we have visibility into. The value originated as a cutoff search, and it remains a cutoff value. The triangulation confirms the framework's $(g_s/ - 2)^n \times \text{textbook}$ structure is self-consistent across multiple observables under a single cutoff; it does not promote that cutoff to a derived quantity. A genuine first-principles derivation — in which r_e emerges from the dual-Dirac equation's internal structure rather than being fit to data — would be a genuinely new piece of

theory work. (See Candidate 3 below for a *partial* identification: the triangulated value already has a clean closed-form Schwinger reading that recasts the derivation question into “*why* does the framework specify this cutoff prescription?” rather than “what numerical value should the cutoff have?”)

This note collects the three candidate starting points we have considered for such a derivation, with one to three paragraphs on each describing what the route would look like, what its plausibility appears to be from outside the framework’s internal logic, and what would be required to pursue it. (A fourth candidate we had initially considered — retrieving a derivation step from the original DRQM I §III.D working notebook that did not make it into the published prose — has been ruled out by your 2026-05-25 author guidance that the original cutoff was a numerical search against g_s alone, with no derivation-step working draft to retrieve.) We are **not proposing any of these three** — we defer to your judgment on which (if any) is natural for the framework. The thread is non-urgent: the triangulated value already serves as the campaign’s current-best-refinement at the framework’s precision floor, and a first-principles derivation is a “would-be-nice-to-have” rather than a load-bearing item.

Candidate 1 — Proper-time self-energy integral

The dual theory’s distinguishing feature is the proper-time first-order Dirac structure: the standard Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ is replaced by a proper-time-formulated version in which the dynamical variable is ∂_τ rather than ∂_t . The natural place to look for r_e as a derived quantity is the framework’s analog of the one-loop electron self-energy diagram: the diagram in which a single photon propagator and a single electron propagator close into a loop on the external electron line.

In standard QED, the one-loop self-energy $\Sigma(p)$ has a UV divergence that is absorbed into mass renormalisation via a counterterm. The cutoff procedure (Pauli–Villars, dimensional regularisation, lattice cutoff) introduces a regulator scale, and in the renormalised expression that scale’s dependence cancels modulo the running of α . In the dual-theory’s proper-time formulation, the analog calculation would feature the proper-time propagator and a proper-time photon propagator. The UV behavior might be qualitatively different from standard QED: possibly finite, possibly with a different divergence structure rooted in the proper-time integral’s analytic properties.

Concretely, the route would set up a Schwinger-like proper-time integral representation of the dual-Dirac propagator, compute the one-loop self-energy with the proper-time photon propagator, and identify the natural scale at which the mass counterterm equals the physical-minus-bare mass difference. If the integral is finite, that scale is determined unambiguously; if divergent, it is set by the renormalisation condition. The output r_e/r_0 would be a numerical function of α and the dual-Dirac structure constants (b, c , the projection-operator coefficients), evaluating to a definite numerical value the campaign could compare against 0.499 420 509 ...

Candidate 2 — Variational determination via the renormalised dual-Dirac equation

In the dual-Dirac equation $\hat{H}_{\text{dual}}\psi = E\psi$, the cutoff r_e appears as a parameter — whether as a hard cutoff in the radial integration, or as a regulator in the small- r behavior of the wavefunction. The eigenvalue E depends on r_e : $E = E(r_e)$. A variational determination would fix r_e by demanding

that the physical electron mass eigenvalue be recovered at the cutoff, $E(r_e) = m_e c^2$ (including any binding-energy or self-energy contributions specified by the framework).

This is analogous to how the standard-QED running coupling $\alpha(\mu)$ is fixed by the renormalisation condition $\alpha(\mu_0) = \alpha_{\text{phys}}$ at a chosen reference scale: a framework parameter is fixed by a physical observable evaluated at the framework’s natural scale. The advantage of this route over Candidate 1 is that it side-steps the UV-divergence question entirely — the variational principle operates within the framework’s renormalised structure, not on the divergent bare-vertex calculation. The disadvantage is that mass-condition-alone might not uniquely determine r_e : additional consistency conditions (gauge invariance of the photon propagator at the cutoff, current conservation at the radial boundary, perhaps the magnetic-moment relation closing the system) may be needed for closure.

If the right set of consistency conditions can be assembled, the variational route is the most direct path from the dual-Dirac equation’s internal structure to a numerical r_e/r_0 . The calculation would be a coupled eigenvalue / renormalisation-condition problem solvable symbolically (Mathematica) once the conditions are specified. The output is again comparable against the triangulated 0.499 420 509 ... — but the route’s credibility depends on whether the additional consistency conditions are framework-internally motivated or look ad hoc.

Candidate 3 — The triangulated value already has a clean closed-form Schwinger reading

The triangulated $r_e/r_0 = 0.499\,420\,509\,912\,831\,7$ is, to within $\sim 10^{-6}$, the closed-form value

$$r_e/r_0 = \frac{2 - \alpha/(2\pi)}{4 + \alpha/\pi} = 0.499\,419\,632\,156\dots$$

obtained by inverting $g_r = 2[1 - 4r_0/(2r + r_0)]$ against the **Schwinger one-loop QED anomalous moment** $g_e^{(1\text{-loop})} = -2 - \alpha/\pi = -2.002\,322\,819\,465\,7\dots$. The residual discrepancy between the closed-form 0.499 419 632 and the triangulated 0.499 420 510 is exactly the gap between the Schwinger one-loop and the all-orders measured $g_e = -2.002\,319\,304\,362\,5\dots$ (which includes the Karplus–Kroll two-loop and higher corrections at $\sim 10^{-6}$ relative). In other words: at one-loop QED precision, the cutoff prescription $r_e = (2 - \alpha/(2\pi))/(4 + \alpha/\pi) \cdot r_0$ is what makes the formula $g_r = 2[1 - 4r_0/(2r + r_0)]$ reproduce the Schwinger result exactly.

This is, from outside the framework’s internal logic, the cleanest and most natural reading of the cutoff identification: the DRQM I §III.D formula is, at the precision the framework’s apparatus delivers, an algebraic re-encoding of the Schwinger one-loop anomalous moment through a particular cutoff substitution. The reason the triangulation across six observables returns essentially the same value as the uni-observable back-fit against g_s alone is that **every other observable’s prediction is $(g_s/ - 2)^n \times \text{textbook}$** , so substituting the measured g_s is the same operation whether done once or six times — and the measured g_s is, to one-loop QED precision, exactly what the closed-form cutoff is engineered to reproduce.

What the *first-principles* derivation question reduces to, on this reading, is: **why does the framework specify this particular cutoff prescription?** Two sub-questions: (i) Is there a derivation in which the dual-Dirac equation’s renormalisation structure produces $r_e = (2 -$

$\alpha/(2\pi))/(4 + \alpha/\pi) \cdot r_0$ as the natural mass-renormalisation scale (Candidates 1 or 2 applied at one-loop precision)? Or (ii) is the cutoff a structural constant of the dual representation itself, such that the combination $2 - \alpha/(2\pi))/(4 + \alpha/\pi)$ comes out of the framework’s algebra without reference to QED at all? Question (i) is a one-loop calculation that should be tractable if the framework’s renormalisation prescription is fully specified; question (ii) requires identifying the framework-internal origin of an α -dependent ratio, which is harder. If question (i) is the natural reading, the campaign can attempt the calculation directly; if question (ii), we would need your guidance on which structural decomposition produces α -dependent ratios in the framework.

We did not see this identification in the published DRQM I §III.D text. If it is a known property of the framework, please tell us — that would simplify the disposition considerably. If it is a coincidence at the campaign’s precision (i.e., the triangulated value happens to be close to the Schwinger closed-form but is not engineered to be), the Karplus–Kroll-level residual is the test: with more precision-spectroscopy observables in the joint fit, the residual should not systematically point at the two-loop QED corrections unless the cutoff genuinely is encoding QED.

Progress update (2026-05-26 PM) — three parallel research branches

Since the morning version of this note went out, three parallel research branches (issues [#64](#), [#65](#), [#66](#) under master [#67](#)) have begun attempting the three candidates. Per Crocco compliance the work below is *substantive AI* with ‘markers throughout the per-branch state logs; the headline findings are summarised here for your morning read. Full per-iteration state logs (math, Wolfram MCP outputs, blockers) are committed under `dev/research/STATE.md`’ on each branch.

Candidate 1 (Proper-time self-energy, [#64](#)) — *live; conceptual re-framing of r_e + baseline scaffolded*

Four iterations: read DRQM I §II–III, *The Classical Electron Problem*, and the Bethe–Salpeter Lamb-shift document; scaffolded `Roadmapping/Mathematica_Notebooks/Quantum_Mechanics/r_e_derivation_self_e` with Cell 1 (baseline standard-QED on-shell $\delta m/m = (3\alpha/(4\pi))[\log(\Lambda^2/m^2) + 1/2]$, Bjorken–Drell Eq. 10.59, verified by Wolfram MCP).

The headline finding is **conceptual rather than computational**: the existing Lamb-shift calculation uses the textbook non-relativistic Bethe (1947) UV cutoff $K \sim mc^2 \Leftrightarrow \lambda_C = \hbar/(mc)$, while the triangulated $r_e \sim r_0/2 = (\alpha/2)\lambda_C$ — **parametrically smaller than the Bethe cutoff by $\alpha/2 \approx 3.65 \times 10^{-3}$** . So r_e is not a UV-loop cutoff at all; it sits at the *Coulomb-binding* scale, not the Compton scale. Physically, at $r = r_0/2$, the Coulomb potential $|V_0| = 2mc^2$ — exactly the pair-production threshold. The natural re-reading: r_e marks the radius inside which the bound-state wavefunction picks up virtual e^+e^- contributions and the §III.D small-component elimination $\psi_2 = c(\boldsymbol{\sigma} \cdot \boldsymbol{\pi})\psi_1/(\lambda - V_0 + mc^2)$ stops being a valid approximation.

Technical direction: the (II.3) “potential-in-the-mass” form is the cleanest kernel — no $V\boldsymbol{\alpha} \cdot \boldsymbol{\pi}/(mc)$ or $\boldsymbol{\alpha} \cdot \nabla V/(mc)$ pieces (those arise from operator non-commutativity in the (II.1) Dirac form), so the kernel reduces to Pauli kinetic + $V^2/(2mc^2)$.

Heuristic sanity check (iter 4). The naive identification $\Lambda = \hbar/(r_e c)$ at $r_e/r_0 = 0.5$ gives $\log(\Lambda^2/m^2) = 11.23$ and $\delta m/m = 2.04 \times 10^{-2}$, vs the natural one-loop coupling $\alpha/(4\pi) = 5.81 \times 10^{-4}$ — a $35\times$ overestimate. Two *distinguishable* resolutions queued for Cells 2–3: (i) the framework

supplies $\Lambda \sim \hbar/(br_e)$ with $b > c$ for bound states (suppresses the log), or (ii) the Bethe-log replacement $\log(K/\langle\Delta E\rangle)$ at fixed photon-loop measure (sum-over-states log, parametrically smaller). Distinguishable because (i) shifts r_e/r_0 via $\log(b/c)$ at fixed $\langle p^2 \rangle$, while (ii) shifts r_e/r_0 via the Bethe-log replacement at fixed photon-loop measure. **Branch status: live, on plan. Outcome-matrix branch still open; Cells 2–3 expected to discriminate (i)/(ii) and produce a numerical r_e/r_0 .**

Candidate 2 (Variational route, #65) — *HALTED at Outcome D: the published NR-expansion K_D is structurally inadequate to pin r_e*

Four iterations enumerated seven closure conditions (only the mass-renormalisation condition #7 was both *framework-internal* and *potentially of sufficient precision*), then pursued it under the textbook working hypothesis $\Delta E_{\text{bind}}^{\text{framework}} = \langle V_0 \rangle$, $\Delta E_{\text{SE}}^{\text{framework}} = 0$, with trial $\psi_1 = Ne^{-r/aa}$ on cutoff-domain $[r_e, \infty)$. The closure equation (Wolfram MCP, dimensionless $\hat{a} = aa/r_0$, $\hat{r}_e = r_e/r_0$):

$$E_{\text{dim}}(\hat{a}, \hat{r}_e) = \frac{1}{2\alpha^2 \hat{a}^2} - \frac{\hat{a} + 2\hat{r}_e - 1}{\hat{a}^2 + 2\hat{a}\hat{r}_e + 2\hat{r}_e^2} = 0.$$

The diagnostic table (Wolfram MCP, $\alpha = 1/137.035999$) shows the **structural inadequacy**:

Regime	$\hat{a}$	$\hat{r}_e$	E_{dim}	Verdict
Electron-radius scale	1	0.5	+9389.0	Kinetic dominates by 4 orders; NR expansion invalid
Bohr scale	$1/\alpha^2 \approx 18\,778$	0.5	-2.662×10^{-5}	Matches $-\alpha^2/2$; cutoff invisible
Bohr scale, no cutoff	$1/\alpha^2$	0	-2.662×10^{-5}	Identical to $\hat{r}_e = 0.5$ to 10 sig figs

At the cutoff scale, both expansion parameters of (III.4) — $V_0/(mc^2)$ and $\hbar/(mcr)$ — are $O(1)$, so the expansion is invalid. At the Bohr scale where the expansion *is* valid, the trial mass-density $r^2 e^{-2r/aa}$ peaks at $r \sim aa \sim 18\,000r_0$, suppressing the cutoff coupling by α^2 . **There is no scale at which both (a) the NR expansion is valid AND (b) the cutoff couples meaningfully to the closure**, so the published expanded K_D cannot pin r_e . **Branch HALTED, Outcome D, BLOCKED.** Two specific clarifications would unblock: (1) a framework-internal $\Delta E_{\text{SE}}^{\text{framework}}(r_e)$ at the cutoff scale, or (2) redirect to the *un-expanded* full H_D under a radial-cutoff regulator (a 5–10 iteration arc, distinct from Route X as defined; not initiated without author endorsement). See Question 2 below.

Candidate 3 (Closed-form Schwinger, #66) — *HALTED at Outcome C-as-published: the $KK + LR + KF$ residual match is algebraically forced by the back-fit, not evidence of intentional encoding*

Four iterations executed the empirical-test path. At 20-digit Wolfram precision with $\alpha = 7.297\,352\,569\,3 \times 10^{-3}$:

Quantity	Value
r_e/r_0 closed-form $(2 - \alpha/(2\pi))/(4 + \alpha/\pi)$	0.499 419 632 156 99
r_e/r_0 triangulated (PR #62)	0.499 420 509 912 83
$\Delta g_e^{\text{obs}} = g_e^{\text{meas}} - g_e^{\text{Schwinger}}$	$+3.5151 \times 10^{-6}$
Karplus–Kroll two-loop $+2C_2(\alpha/\pi)^2$,	$+3.5446 \times 10^{-6}$
$C_2 = 0.328\,478\,965\,579$	
Laporta–Remiddi three-loop $-2C_3(\alpha/\pi)^3$,	-2.961×10^{-8}
$C_3 = 1.181\,241\,456\,587$	
Kinoshita–Fukuda four-loop $+2C_4(\alpha/\pi)^4$,	$+1.1 \times 10^{-10}$
$C_4 \approx 1.9106$	
Sum (KK + LR + KF)	$+3.51511 \times 10^{-6}$
Residual: $\Delta g_e^{\text{obs}} - \Delta g_e^{\text{pred, all}}$	-9.7×10^{-12}

The verdict. Iter 3 read DRQM I §III.D Eqs. (III.18)–(III.23) line-by-line and confirmed that **the as-published §III.D derivation does *not* produce r_e/r_0 as a closed form in α** : Eq. (III.22)’s numerical value $r_e = 0.499857150068631 \cdot r_0$ is presented as an empirical fact (the uni-observable numerical search you confirmed on 2026-05-25), not derived. Consequently the 10^{-11} agreement above is **algebraically forced** by the back-fit definition of r_e^{triang} : any cutoff satisfying $g_r(r) = g_e^{\text{meas}}$ will, when subtracted from the closed-form one-loop value, yield via dg_r/dr propagation a Δg_e identical to the all-orders-QED-beyond-one-loop content of measured g_e — i.e., to KK + LR + KF. The KK-consistency observation is *necessary* but *not sufficient* for intentional encoding, and the as-published apparatus does not supply the sufficient piece.

Branch HALTED, Outcome C-as-published. Finding 2 stays [warn] CHARACTERISED. The canonical record is now committed as a new subsection of [Equation_Verification/Dual_Relativ](#) §III.D (“Schwinger identification — empirical residual test”). Path to outcome B (intentional Schwinger encoding $\Rightarrow$ Finding 2 [OK]) now passes solely through a first-principles rederivation that *produces* the closed-form as a derived identity — i.e., through Candidate 1 if it lands on the closed form, or through a sub-route of Candidate 2 if author input redirects to the un-expanded full H_D .

Consolidated questions for you

After this overnight run, two questions remain load-bearing (the third has been resolved by the iter-3 §III.D line-by-line read). Even one answer accelerates the remaining live branch:

1. **(Candidate 1, live branch)** Does the framework specify a proper-time photon propagator form? Two natural candidates surface: (i) Schwinger proper-time with b replacing c in the dispersion, $k^2 = (\omega/b)^2 - \mathbf{k}^2$; (ii) standard Feynman propagator with the b/c conversion absorbed into the source coupling rather than the propagator. The two give different numerical r_e/r_0 at the same order in α , and the iter-4 sanity check shows the difference is at the $\log(b/c)$ level — discriminable.
2. **(Candidate 2, HALTED $\rightarrow$ unblocking)** The published expanded K_D of (III.4) is structurally inadequate to pin r_e , as the diagnostic table above shows. Two unblocking moves: **(2a)** does the framework specify a $\Delta E_{\text{SE}}^{\text{framework}}(r_e)$ at the cutoff scale that we should have been carrying in $\langle K_D \rangle = m_e c^2 + \Delta E_{\text{bind}} + \Delta E_{\text{SE}}$? Or **(2b)** is the variational determination intended to operate on the *un-expanded* full H_D under a radial-cutoff regulator (radial-Dirac eigenvalue problem on $r \in [r_e, \infty)$) rather than the expanded K_D of (III.4)? The un-expanded route is a 5–10 iteration arc we have not begun without your endorsement.

The original Question 3 from earlier today (does §III.D derive r_e/r_0 as a closed form in α ?) was answered by the iter-3 line-by-line read: **as published, §III.D does not derive a closed form** — the numerical value is an empirical input from your uni-observable numerical search. The remaining open piece on the closed-form-encoding question is whether an in-progress or planned rederivation (per Candidate 1, or a redirected Candidate 2) would *produce* the closed form. That is now the substantive thread for the master tracker #67.

Closing — how this thread sits

We expect the most likely outcome, given your 2026-05-25 guidance and the campaign’s posture, is that none of these three candidates rises above the triangulated cutoff in usefulness for the campaign’s current scope. The triangulated value is empirically well-constrained at the framework’s precision floor (the joint-fit consistency across six manifestations of the $(g_s/-2)^n \times$ textbook scaling), agrees with measurement at that floor, and is sufficient for the campaign’s downstream work without a first-principles derivation behind it. The Scope 1 thread is therefore non-urgent — it remains open as a “would-be-nice-to-have” for future framework development, not as a load-bearing item. Candidate 3’s Schwinger closed-form identification — if it is intentional in the framework’s construction — may already be the answer.

If one of the three candidates does look natural to you, we would be glad to pursue it via Mathematica symbolic calculation with your guidance on the framework’s internal logic. If none of them looks natural, the triangulated value stands as the campaign’s r_e disposition and Scope 1 can be closed without a derivation. Either way, our position is that the triangulation has confirmed the $(g_s/-2)^n \times$ textbook structure is self-consistent under a single cutoff at the framework’s precision floor; that is the campaign’s current honest-scope position on the r_e question.

The *Progress update* above adds three substantive pieces of physical content on top of the morning version. First, the **re-framing of r_e** from “UV cutoff” to “Coulomb-binding scale at the pair-production threshold” (Candidate 1) — a statement about what r_e *is* in the framework’s small-component-elimination apparatus, independent of which derivation route eventually settles its first-principles status, and worth your read even if Scope 1 closes without a derivation. Second, the **structural inadequacy of the published expanded K_D for variational r_e determination** (Candidate 2 -> Outcome D) — at no scale do (a) the validity of the (III.4) NR expansion and (b) meaningful coupling of the cutoff to the closure equation coexist. Third, the **algebraically-forced character of the Schwinger closed-form agreement** (Candidate 3 -> Outcome C-as-published) — the 10^{-11} KK + LR + KF residual match is necessary but not sufficient for intentional encoding, and the as-published §III.D does not provide the sufficient piece. The disposition of Scope 1 now turns on the live Candidate 1 branch (and on whether Candidate 2 redirects to the un-expanded full- H_D arc with your endorsement).

— Trey